ABinterface: Short-Time Coherent A/B Interface Model

Theoretical Background, Hamiltonian Definition, Time Propagation, and User Guide

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1 Purpose and Physical Scope

ABInterface implements a compact one-particle density-matrix toy model for short-time laser-driven carrier and spin redistribution at a non-matching interface between two materials, denoted by A and B. The model is designed for qualitative mechanism analysis rather than quantitative materials prediction. It is useful when one wants to test whether a set of coherent elementary couplings, including optical excitation, spin-orbit-induced spin mixing, and interlayer hopping, can generate a target projected-occupation pattern. In such a short-time electronic-dynamics picture, the laser excites the electronic system and, after the laser pulse has vanished, the electronic density matrix continues to evolve coherently under the field-free Hamiltonian. Phenomenological electron-phonon relaxation, Boltzmann collision terms, Lindblad baths, or manually imposed downhill rates are not part of the present code.

The code does not track a tagged electron and does not impose a single classical trajectory. A chain-like pattern such as

$$A_v \uparrow \xrightarrow{\text{laser}} A_c \uparrow \xrightarrow{\text{SOC mixing}} A_c \downarrow \xrightarrow{\text{interlayer hopping}} B_c \downarrow \xrightarrow{\text{changes in the } B_v\downarrow \text{ projected occupation}} B_v \downarrow . \tag{1}$$

should be interpreted only as an emergent description of projected occupations, *not* hard-coded as a single ordered path. The implemented model contains coherent elementary Hamiltonian couplings: optical excitation, material-internal SOC mixing, and interlayer hopping.

2 Basis

Each material and spin block contains N single-particle states. The total Hilbert-space dimension is

$$D = 4N. \tag{2}$$

The basis ordering is

$$A_\uparrow, \quad A_\downarrow, \quad B_\uparrow, \quad B_\downarrow. \tag{3}$$

In each block, the first $N/2$ states are valence-like and the last $N/2$ states are conduction-like. Therefore N must be even. In index form,

Block	Index range	Meaning
$A_\uparrow$	$0, \ldots, N-1$	material A, spin up
$A_\downarrow$	$N, \ldots, 2N-1$	material A, spin down
$B_\uparrow$	$2N, \ldots, 3N-1$	material B, spin up
$B_\downarrow$	$3N, \ldots, 4N-1$	material B, spin down

Within any block,

$$0, \ldots, N/2-1 \quad \text{are valence-like states,} \tag{4}$$

while

$$N/2, \ldots, N-1 \quad \text{are conduction-like states.} \tag{5}$$

A generic basis state is denoted

$$|X, b, m, s\rangle , \tag{6}$$

where $X \in \{A, B\}$ is the material label, $b \in \{v, c\}$ is the valence or conduction manifold, m is a local band-ladder index, and $s \in \{\uparrow, \downarrow\}$ is the spin label. The index m is not a magnetic quantum number; it is simply the discrete level index used by the toy model.

3 Hamiltonian Overview

The time-dependent density matrix is denoted by $\rho(t)$, and the time-dependent Hamiltonian is

$$H(t) = H_{\text{static}} + H_{\text{laser}}(t), \tag{7}$$

where

$$H_{\text{static}} = H_{\text{bands}} + H_{\text{SOC,mix}} + H_{\text{interlayer}}. \tag{8}$$

During the laser pulse, $H_{\text{laser}}(t) \neq 0$. After the pulse, $H_{\text{laser}}(t) = 0$, and the density matrix continues to evolve coherently under H_{static}.

The model contains the following physical components:

1. **No-SOC non-matching bands**: materials A and B have independent gaps, energy offsets, and band widths.

2. **SOC-derived diagonal splitting**: the diagonal spin splitting is controlled by the signed material-internal parameters $\lambda_{\text{soc},A}$ and $\lambda_{\text{soc},B}$.

3. **Material-internal SOC mixing**: off-diagonal couplings mix $\uparrow$ and $\downarrow$ components within the same material and band manifold.

4. **Spin-label-conserving interlayer hopping**: the default interlayer hopping couples $A_c s \leftrightarrow B_c s$, with optional $A_v s \leftrightarrow B_v s$.

5. **Laser excitation**: The laser field couples valence-like and conduction-like levels within each material. In the chosen diabatic basis, the explicit optical matrix element does not contain a spin-flip term. However, the full laser-driven dynamics is not generally spin-conserving, because of the SOC-mixing terms.

6. **Post-pulse processes**: Density matrix continues to evolve coherently in a short-time regime. Carrier redistribution includes material-internal SOC mixing and interlayer hopping. No phenomenological incoherent relaxation, downhill rate, or collision term is included.

Important Modeling Clarification

Therefore statements such as

$$A_v \uparrow \to A_c \uparrow \to A_c \downarrow \to B_c \downarrow \to B_v \downarrow \tag{9}$$

should be read as an *effective pathway interpretation* of projected occupation changes, not as a hard-coded algorithmic transition chain. Directly computed observables are occupations, coherences, and their projections onto material, spin, band, and static-eigenstate subspaces.

4 No-SOC Band Energies

The no-SOC band structure is specified by material-dependent gaps, offsets, and bandwidths. For material $X \in \{A, B\}$, the valence-band maximum and conduction-band minimum before SOC splitting are approximately

$$E^{(0)}_{X,\text{VBM}} = \text{offset}_X - \frac{1}{2}\text{gap}_X, \tag{10}$$

$$E^{(0)}_{X,\text{CBM}} = \text{offset}_X + \frac{1}{2}\text{gap}_X. \tag{11}$$

The valence ladder is constructed downward from the VBM:

$$E^{(0)}_{X,v,m} = E^{(0)}_{X,\mathrm{VBM}} - \epsilon^{X,v}_m, \tag{12}$$

where $\epsilon^{X,v}_m$ spans the valence bandwidth. The conduction ladder is constructed upward from the CBM:

$$E^{(0)}_{X,c,n} = E^{(0)}_{X,\mathrm{CBM}} + \epsilon^{X,c}_n, \tag{13}$$

where $\epsilon^{X,c}_n$ spans the conduction bandwidth.

5 SOC-Derived Diagonal Splitting

The code does not expose independent spin-splitting parameters. Instead, the diagonal SOC splitting is derived from material SOC strengths:

$$\lambda_{\mathrm{soc},A}, \qquad \lambda_{\mathrm{soc},B}. \tag{14}$$

The convention used by the code is

$$E_{\uparrow} = E_0 + \frac{\Delta}{2}, \qquad E_{\downarrow} = E_0 - \frac{\Delta}{2}. \tag{15}$$

The diagonal SOC splitting is signed and is taken directly from the input parameters:

$$\Delta_A = \lambda_{\mathrm{soc},A}, \qquad \Delta_B = \lambda_{\mathrm{soc},B}. \tag{16}$$

The code uses the convention

$$E_{X,\uparrow} = E^{(0)}_X + \frac{\Delta_X}{2}, \qquad E_{X,\downarrow} = E^{(0)}_X - \frac{\Delta_X}{2}. \tag{17}$$

Therefore a negative $\lambda_{\mathrm{soc},X}$ places the spin-down level above the spin-up level, while a positive $\lambda_{\mathrm{soc},X}$ places the spin-up level above the spin-down level. The default values are negative.

6 Hamiltonian Terms

6.1 Band Hamiltonian

The diagonal band Hamiltonian is

$$H_{\mathrm{bands}} = \sum_i E_i \left|i\right\rangle \left\langle i\right|, \tag{18}$$

where $|i\rangle$ is a diabatic basis state such as $|A, c, n, \uparrow\rangle$ or $|B, v, m, \downarrow\rangle$. The diagonal energies E_i are obtained from (17).

6.2 Material-Internal SOC Mixing

Diagonal SOC splitting alone does not transfer population between spin channels. To realize the step like

$$A_c \uparrow \to A_c \downarrow, \tag{19}$$

the model includes off-diagonal material-internal SOC mixing. In the simplest same-band-index form,

$$H_{\mathrm{SOC,mix}} = \sum_{X=A,B} \sum_{b=v,c} \sum_m \Omega_{X,b} \left|X, b, m, \downarrow\right\rangle \left\langle X, b, m, \uparrow\right| + \mathrm{h.c.} \tag{20}$$

Here $X \in \{A, B\}$ is the material label, $b \in \{v, c\}$ is the band-manifold label, and m is the local band-ladder index inside that material and band manifold. The index m is not a magnetic quantum number; it is simply the discrete level index of the toy-model valence or conduction ladder. For each fixed X, b, and m, this term couples the spin-up-like and spin-down-like components of the same local band-ladder state.

The coefficient $\Omega_{X,b}$ is the material- and band-dependent SOC spin-mixing amplitude. In the present implementation, it is independent of m within a given (X, b) block. A more detailed model could use an m-dependent coefficient $\Omega_{X,b,m}$, but that is not part of the current code.

The corresponding software parameters are

Parameter	Meaning
`soc_mix_cb_A`	$\Omega_{A,c}$, A conduction-band up/down SOC mixing
`soc_mix_cb_B`	$\Omega_{B,c}$, B conduction-band up/down SOC mixing
`soc_mix_vb_A`	$\Omega_{A,v}$, A valence-band up/down SOC mixing
`soc_mix_vb_B`	$\Omega_{B,v}$, B valence-band up/down SOC mixing

The conduction-band mixing parameters are especially important because they allow the elementary spin-mixing step like

$$A_c \uparrow \to A_c \downarrow . \tag{21}$$

They also allow SOC-mixed B_c states, so occupation accumulated in $B_c \downarrow$ can indirectly affect $B_c \uparrow$ optical availability through mixed eigenstates.

6.3 Spin-Label-Conserving Interlayer Hopping

The interlayer hopping matrix is diagonal in the diabatic spin label. The code does not implement a direct interlayer spin-flip term of the form $A_c \uparrow \leftrightarrow B_v \downarrow$. Instead, it allows hopping between states with the same spin and the same band-manifold type:

$$A_{b,m,s} \leftrightarrow B_{b,n,s}, \qquad b \in \{c, v\}, \qquad s \in \{\uparrow, \downarrow\}. \tag{22}$$

Here $b = c$ denotes conduction-manifold hopping and $b = v$ denotes valence-manifold hopping. The indices m and n are local band-ladder indices in materials A and B, respectively. They are not magnetic quantum numbers. They label the discrete levels inside the toy-model valence or conduction ladder.

The general same-manifold interlayer Hamiltonian can be written as

$$H_{\text{interlayer}} = \sum_{b\in\{c,v\}} \sum_{m,n,s} T^{(b)}_{mn,s} \left|B, b, n, s\right\rangle \left\langle A, b, m, s\right| + \text{h.c.} \tag{23}$$

The hopping matrix element is modeled as

$$T^{(b)}_{mn,s} = t^{(b)}_{AB} F^{(b)}_{\text{band}}(m, n) F^{(b)}_{E}(m, n, s). \tag{24}$$

The band-dependent hopping scale is

$$t^{(c)}_{AB} = \texttt{tAB}, \qquad t^{(v)}_{AB} = \texttt{tAB\_vv}. \tag{25}$$

Thus `tAB` controls conduction-conduction hopping,

$$A_{c,m,s} \leftrightarrow B_{c,n,s}, \tag{26}$$

whereas `tAB_vv` controls valence-valence hopping,

$$A_{v,m,s} \leftrightarrow B_{v,n,s}. \tag{27}$$

The default value is tAB_vv $= 0$, so the default model mainly uses conduction-conduction interlayer hopping. Nevertheless, the code structure is not restricted to $c-c$ hopping; valence-valence hopping is present as an optional same-spin, same-manifold channel.

The band-index filter is

$$F_{\text{band}}^{(b)}(m,n) = \exp\left[-\frac{\left(d_{b,m}^{A} - d_{b,n}^{B}\right)^2}{2w_{\text{inter}}^2}\right]. \tag{28}$$

Here

$$w_{\text{inter}} = \texttt{interface\_band\_width}. \tag{29}$$

The quantities $d_{b,m}^{A}$ and $d_{b,n}^{B}$ are distances from the relevant band edge, measured in band-index space. For conduction states,

$$d_{c,m}^{X} = 0 \tag{30}$$

means the CBM-like state of material X, while $d_{c,m}^{X} = 1, 2, \ldots$ denote higher conduction states farther from the CBM. For valence states,

$$d_{v,m}^{X} = 0 \tag{31}$$

means the VBM-like state of material X, while $d_{v,m}^{X} = 1, 2, \ldots$ denote deeper valence states farther below the VBM. Therefore, interface_band_width controls how selectively the interface couples band-edge states to corresponding band-edge states. A small value makes the hopping strongly concentrated between states with similar distance from the relevant band edge, while a large value allows more remote band-ladder states to couple.

The energy-mismatch filter is

$$F_{E}^{(b)}(m,n,s) = \exp\left[-\frac{\left(\Delta E_{mn,s}^{(b)}\right)^2}{2\sigma_{\text{hyb}}^2}\right], \tag{32}$$

with

$$\sigma_{\text{hyb}} = \texttt{hybrid\_energy\_width}, \tag{33}$$

and

$$\Delta E_{mn,s}^{(b)} = E_{A,b,m,s} - E_{B,b,n,s}. \tag{34}$$

Thus $\Delta E_{mn,s}^{(b)}$ is the energy mismatch between the two same-spin states that are being coupled across the interface. A small hybrid_energy_width suppresses hopping between energetically mismatched A/B states, while a large value makes the interface hopping less sensitive to this mismatch.

Combining these definitions, the explicit matrix element is

$$T_{mn,s}^{(b)} = t_{AB}^{(b)} \exp\left[-\frac{\left(d_{b,m}^{A} - d_{b,n}^{B}\right)^2}{2\,\texttt{interface\_band\_width}^2}\right] \exp\left[-\frac{\left(E_{A,b,m,s} - E_{B,b,n,s}\right)^2}{2\,\texttt{hybrid\_energy\_width}^2}\right]. \tag{35}$$

The correspondence between the formula and the user-facing parameters is therefore:

Symbol	Input parameter or code-generated quantity
$t_{AB}^{(c)}$	tAB
$t_{AB}^{(v)}$	tAB_vv
w_{inter}	interface_band_width
σ_{hyb}	hybrid_energy_width
$d_{b,m}^{X}$	band-edge distance of state (X, b, m), generated from the local band index
$\Delta E_{mn,s}^{(b)}$	$E_{A,b,m,s} - E_{B,b,n,s}$, generated from the spin-resolved diagonal energies

This interlayer Hamiltonian is part of the static Hamiltonian and therefore contributes to the unitary, coherent part of the dynamics.

7 Laser Pulse and Optical Coupling

The laser is represented by an intra-material coupling between valence-like and conduction-like levels. In the chosen diabatic basis, the explicit optical matrix element is diagonal in the spin label:

$$X_v s \leftrightarrow X_c s, \qquad X \in \{A, B\}, \quad s \in \{\uparrow, \downarrow\}. \tag{36}$$

This does not imply conservation of the spin projection during the full time evolution, because $H_{\mathrm{SOC,mix}}$) is present during and after the pulse. The laser Hamiltonian is

$$H_{\mathrm{laser}}(t) = F(t) \sum_{X,m,n,s} d_{mn}^X \left| X, c, n, s \right\rangle \left\langle X, v, m, s \right| + \mathrm{h.c.} \tag{37}$$

7.1 Pulse Shape

In full-carrier mode,

$$F(t) = A_0 \sin^2\left(\frac{\pi t}{T_{\mathrm{pulse}}}\right) \cos\left(\frac{\omega t}{\hbar}\right), \qquad 0 < t < T_{\mathrm{pulse}}. \tag{38}$$

Outside the pulse window, $F(t) = 0$. In RWA/envelope mode, the rapidly oscillating carrier factor is removed. The drive is still time-dependent and follows the pulse envelope, $F(t) = A_0 \sin^2(\pi t/T_{\mathrm{pulse}})$.

7.2 Optical Matrix Elements

The optical matrix element is modeled as

$$d_{mn}^X = d_0^X F_{\mathrm{res}}(E_c - E_v) F_{\mathrm{edge}}(m, n). \tag{39}$$

The resonance factor is

$$F_{\mathrm{res}} = \exp\left[-\frac{(E_c - E_v - \hbar\omega)^2}{2\sigma_{\mathrm{opt},X}^2}\right]. \tag{40}$$

The edge-overlap factor uses distances from CBM and VBM. If $d_c = 0$ denotes the CBM and $d_v = 0$ denotes the VBM, then

$$F_{\mathrm{edge}} = \exp\left[-\frac{(d_c - d_v)^2}{2w_{\mathrm{band}}^2}\right]. \tag{41}$$

This definition intentionally couples CBM-like states most strongly to VBM-like states and avoids the earlier unphysical pairing of CBM with the deepest valence states.

7.3 Optical Matrix-Element Formula and Input-Parameter Mapping

Combining the resonance and band-edge overlap factors, the optical matrix element used by the laser term can be written as

$$d_{mn}^X = d_0^X \exp\left[-\frac{(E_{X,c,n,s} - E_{X,v,m,s} - \hbar\omega)^2}{2\sigma_{\mathrm{opt},X}^2}\right] \exp\left[-\frac{(d_{c,n} - d_{v,m})^2}{2w_{\mathrm{band}}^2}\right], \tag{42}$$

where $X = A$ or B. The correspondence between the symbols in this expression and the software inputs is:

Symbol	Input parameter or generated quantity
X	material label, either A or B
d_0^A	`dA0`
d_0^B	`dB0`
$\hbar\omega$	`omega_eV`
$\sigma_{\mathrm{opt},A}$	`optical_energy_width_A`, or fallback `optical_energy_width`
$\sigma_{\mathrm{opt},B}$	`optical_energy_width_B`, or fallback `optical_energy_width`
w_{band}	`band_overlap_width`
E_c, E_v	generated from gaps, offsets, bandwidths, and SOC-derived diagonal splitting
d_c, d_v	generated from local band indices relative to CBM and VBM

For a conduction state, d_c is the distance from the CBM in band-index space:

$$\mathrm{CBM} \to d_c = 0, \qquad \text{next CB} \to d_c = 1, \qquad \text{next CB} \to d_c = 2, \dots \tag{43}$$

For a valence state, d_v is the distance from the VBM:

$$\mathrm{VBM} \to d_v = 0, \qquad \text{next lower VB} \to d_v = 1, \qquad \text{next lower VB} \to d_v = 2, \dots \tag{44}$$

Thus `band_overlap_width` controls how strongly VBM-like states couple to CBM-like states and how quickly this coupling falls off for mismatched band-edge distances.

The state energies E_c and E_v are not direct user inputs. They are generated by the band model from

$$\{\texttt{gap}, \texttt{offset}, \texttt{bandwidth\_v}, \texttt{bandwidth\_c}, \lambda_{\mathrm{soc}}\}. \tag{45}$$

Consequently, increasing `dA0` or `dB0` increases the material-specific optical coupling scale, but it does not by itself guarantee strong excitation. The transition must also be close enough to resonance with `omega_eV` and sufficiently allowed by the band-index overlap factor.

In compact form, the optical coupling is controlled by four distinct ingredients:

Input	Role
`dA0`, `dB0`	material-dependent optical coupling scale
`omega_eV`	photon energy
`optical_energy_width`, `optical_energy_width_A/B`	resonance energy window
`band_overlap_width`	band-index overlap window

8 Initial State

The initial state is not a manually filled diabatic valence configuration. Instead, the software constructs the full static Hamiltonian

$$H_{\mathrm{static}} = H_{\mathrm{bands}} + H_{\mathrm{SOC,mix}} + H_{\mathrm{interlayer}}, \tag{46}$$

diagonalizes it,

$$H_{\mathrm{static}} |\phi_n\rangle = \varepsilon_n |\phi_n\rangle, \tag{47}$$

and fills the lowest $2N$ static eigenstates:

$$\rho(0) = \sum_{n=1}^{2N} |\phi_n\rangle \langle\phi_n|. \tag{48}$$

This avoids a spurious quench at $t = 0$ and ensures that the initial density matrix already includes static interlayer and SOC hybridization.

9 Coherent Time Propagation

During each time step, the Hamiltonian is evaluated at the midpoint:

$$H_{\mathrm{mid}} = H\left(t + \frac{\Delta t}{2}\right). \tag{49}$$

The unitary propagator is

$$U(t + \Delta t, t) = \exp\left[-\frac{ii}{\hbar} H_{\mathrm{mid}} \Delta t\right]. \tag{50}$$

The density matrix is updated by

$$\rho(t + \Delta t) = U\rho(t)U^\dagger. \tag{51}$$

The implementation computes the exponential by diagonalizing the Hermitian midpoint Hamiltonian:

$$H_{\mathrm{mid}} = V\epsilon V^\dagger, \qquad U = V \exp\left[-\frac{i\epsilon\Delta t}{\hbar}\right] V^\dagger. \tag{52}$$

For the small matrices used in this toy model, direct dense diagonalization is transparent and sufficiently fast.

10 Observables and Output Files

The software outputs several time-dependent observables. The most important are:

- material- and spin-projected occupations: $N_{A\uparrow}$, $N_{A\downarrow}$, $N_{B\uparrow}$, $N_{B\downarrow}$;
- selected projected occupations used to diagnose chain-like redistribution: $A_v \uparrow$, $A_c \uparrow$, $A_c \downarrow$, $B_c \downarrow$, $B_v \downarrow$, and $B_c \uparrow$;
- total spin projections $N_\uparrow$ and $N_\downarrow$;
- a magnetic-moment proxy $M = N_\uparrow - N_\downarrow$;
- valence and conduction occupation sums;
- static eigenbasis low/high occupations;
- state-resolved occupation changes relative to pulse end.

The standard output files include

```
*_projected_material_spin.png
*_pathway_projected_occupations.png
*_projected_spin.png
*_projected_magnetic_moment.png
*_projection_vs_eigen_occupation.png
*_observables.csv
*_state_occupation_change_from_pulse_end_levels.png
*_state_occupation_change_from_pulse_end_levels.csv
```

The selected-projection plot is a diagnostic for whether the coherent elementary channels produce a chain-like redistribution pattern in projected occupations. It indicates whether the coherent elementary channels generate a chain-like redistribution pattern.

11 Parameter Reference

11.1 Band Parameters

Parameter	Meaning
N	Number of states per material and spin block. Must be even.
gap_A, gap_B	No-SOC band gaps of materials A and B.
offset_A, offset_B	Rigid energy shifts of the no-SOC spectra.
bandwidth_v_A, bandwidth_v_B	Valence bandwidths.
bandwidth_c_A, bandwidth_c_B	Conduction bandwidths.

11.2 SOC Parameters

Parameter	Meaning
lambda_soc_A, lambda_soc_B	Signed material-internal SOC scales used directly as diagonal spin splitting. Negative values place spin-down above spin-up; positive values place spin-up above spin-down.
soc_mix_cb_A, soc_mix_cb_B	Off-diagonal up/down SOC mixing in conduction bands.
soc_mix_vb_A, soc_mix_vb_B	Off-diagonal up/down SOC mixing in valence bands.

11.3 Interlayer Parameters

Parameter	Meaning
tAB	Spin-label-conserving $A_c s \leftrightarrow B_c s$ hopping scale.
tAB_vv	Optional spin-label-conserving $A_v s \leftrightarrow B_v s$ hopping scale.
interface_band_width	Band-index overlap width for interlayer hopping.
hybrid_energy_width	Energy mismatch width for interlayer hopping.

11.4 Laser Parameters

Parameter	Meaning
A0	Laser amplitude in model units.
omega_eV	Photon energy $\hbar\omega$ in eV.
pulse_duration	Pulse duration in fs.
carrier	Either full or rwa.
dA0, dB0	Material-dependent optical coupling scales d_0^A and d_0^B in the optical matrix element.
optical_energy_width	Fallback optical resonance width $\sigma_{\rm opt}$ used when material-specific widths are not supplied.
optical_energy_width_A, optical_energy_width_B	Material-specific optical widths. Blank or None means use the fallback.

`band_overlap_width`	Band-edge overlap width w_{band} for intra-material matrix elements.

11.5 Time Parameters

Parameter	Meaning
`t_final`	Final simulation time in fs.
`dt`	Time step in fs.
`compare_time_ref`	Reference time for state-resolved difference plots. Default is pulse end.
`compare_time_1`, `compare_time_2`	Left and right comparison times for state-resolved level plots.

11.6 Plotting Parameters

Parameter	Meaning
`plot_mode`	Plot either changes (`delta`) or absolute occupations.
`max_plot_points`	Maximum number of plotted time points. Does not affect propagation.
`delta_color_scale`	Manual state-resolved color scale for Δn. If ≤ 0, an automatic scale is used.
`delta_color_norm`	Either `linear` or `symlog`.
`delta_color_percentile`	Percentile used for automatic color scaling.
`delta_colormap`	Diverging colormap for state-resolved occupation changes.
`level_half_length`, `level_x_jitter`, `level_linewidth`, `level_alpha`	Appearance controls for state-resolved energy-level plots.

12 Command-Line Usage

From the repository root, run without installation by setting `PYTHONPATH`:

```
PYTHONPATH=. python -m ABinterface --help
```

A typical run is

```
PYTHONPATH=. python -m ABinterface \
  --N 12 \
  --pulse-duration 20 \
  --t-final 70 \
  --dt 0.02 \
  --carrier full \
  --omega-eV 2.0 \
  --lambda-soc-A -0.05 \
  --lambda-soc-B -0.05 \
  --soc-mix-cb-A 0.02 \
  --soc-mix-cb-B 0.02 \
  --tAB 0.08 \
  --compare-time-1 22 \
  --compare-time-2 50 \
```

```
  --delta-color-scale 0.08 \
  --delta-color-norm linear
```

13 Desktop GUI Usage

The Tkinter GUI can be launched with

```
PYTHONPATH=. python -m ABinterface.gui
```

or, after installation,

```
ABinterface-gui
```

The GUI uses a tabbed layout:

Tab	Contents
Bands	Gaps, offsets, bandwidths, and basis size
SOC + Interlayer	Signed material SOC splitting, SOC mixing, interlayer hopping
Laser	Pulse, photon energy, optical matrix elements
Time	Final time, Time step, Comparison times
Plotting	Color scales and Figure appearance

The top bar contains the output directory, `Run simulation`, `Reset`, and `Open output`. The `Reset` button restores all parameter widgets to `ModelConfig` defaults.